

Three Pāṇinian mechanisms

Input (English) with kāraka roles

The girl

kartā
agent

gave

verb

the boy

sampradāna
recipient

a book

karman
patient

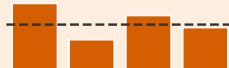
(1) N-hot role embedding



10-dim sparse N-hot

$$W \cdot \mathbf{n} \oplus \mathbf{e}_{\text{tok}}$$

(2) Stratified masking



kartā verb sampr.karman
mean rate matched

(3) śābdabodha head

role classifier
(per token)

discarded at inference

Transformer encoder · 14 L · 768 d · RoPE

$$\mathcal{L}_{\text{MLM}} + \lambda \mathcal{L}_{\text{Aux}}$$